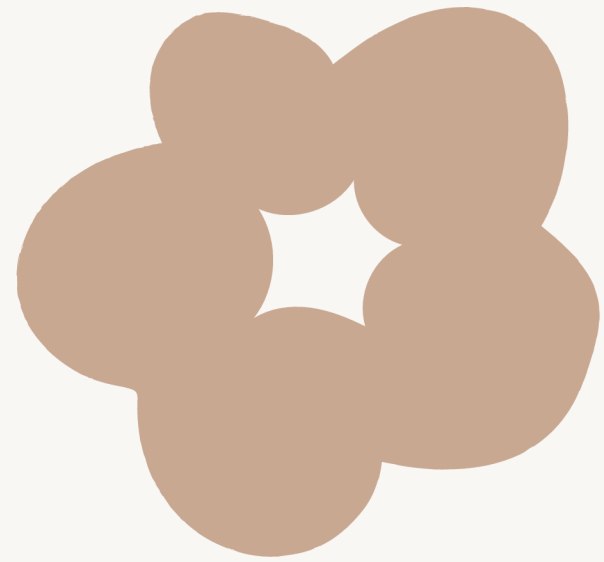




Medical AI Explanation Agent

*MedGemma + Claude Dual-AI
Architecture*

Translating Clinical Reports into Patient-
Friendly Language



The Problem

Medical reports are written for clinicians — not patients.

Clinical Jargon

Terms like 'parapneumonic effusion', 'eGFR 62 mL/min/1.73m²', 'HbA1c 8.2%' are meaningless to most patients — causing anxiety and confusion.

Delayed Understanding

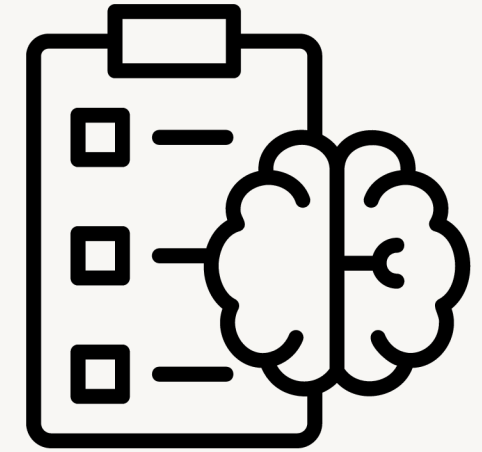
Patients often wait days to speak with a doctor to understand their own test results, delaying critical follow-up actions.

Missed Urgency

Without clear priority signals, patients may ignore HIGH urgency findings — e.g., pneumonia with pleural effusion — thinking they can 'wait and see'.

Project Goal

Build a dual-AI medical explanation system that takes raw clinical reports, extracts clinical insights using a domain-specialist model, and translates them into empathetic, actionable plain English for patients.



COMPARE

MedGemma 4B vs MedGemma 27B on clinical accuracy, speed, and terminology depth

EVALUATE

MedGemma vs Claude on patient communication, urgency detection, and completeness

DEMONSTRATE

The complementary dual-AI pipeline combining domain expertise + communication



Why MedGemma?

A Google-trained, clinically specialized open-weight model.

✗ General LLMs Fall Short

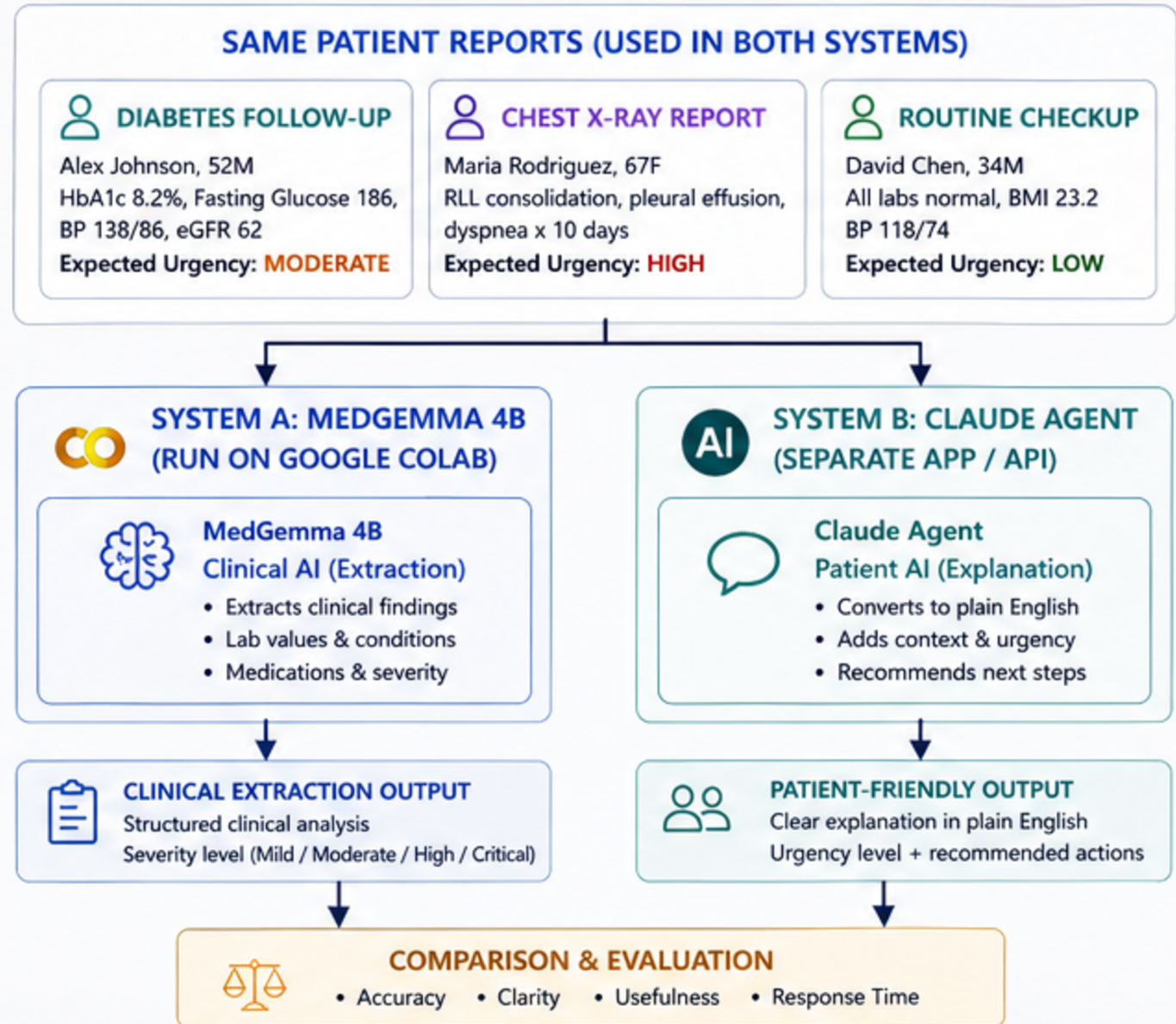
- ✗ No ICD-10 code awareness
- ✗ Hallucinate lab reference ranges
- ✗ Don't use clinical shorthand (BID, QD, eGFR)
- ✗ Not optimized for structured clinical output
- ✗ Generic medical knowledge, not specialized

✓ MedGemma Advantages

- ✓ Trained on medical literature & PubMed
- ✓ Understands ICD-10, clinical acronyms
- ✓ Correctly flags abnormal lab thresholds
- ✓ Free & open-weight — no API cost
- ✓ Available in 4B (fast) and 27B (deep) variants

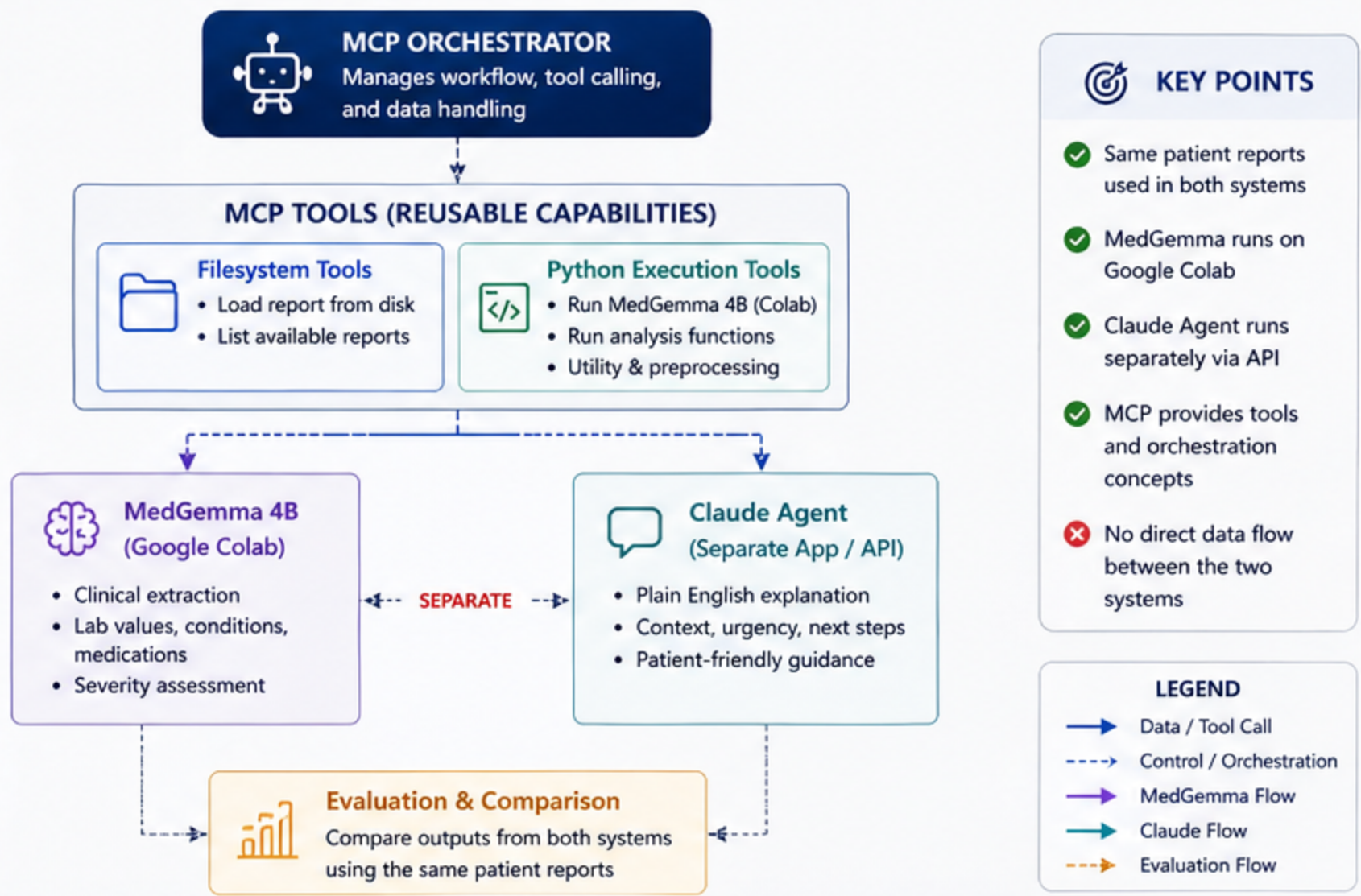
Dual-AI Pipeline Architecture

1 OVERVIEW: SAME INPUT, TWO SEPARATE SYSTEMS

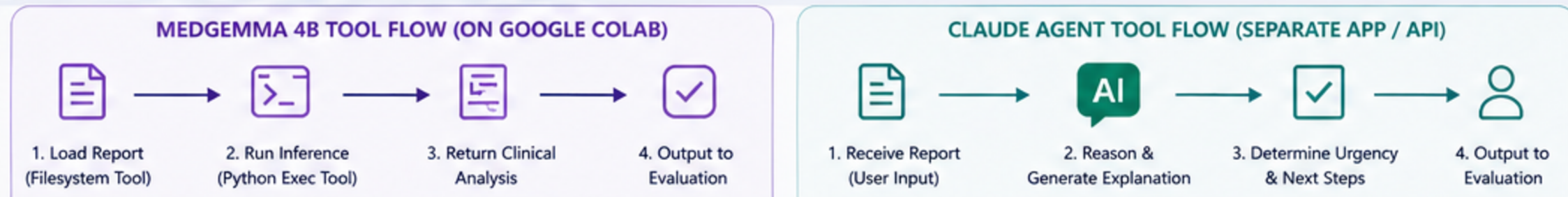


2 MCP ARCHITECTURE (PROJECT DESIGN)

MCP (Model Context Protocol) is the orchestration framework of the project. It provides tools and standard interfaces but does **NOT** connect the two systems directly.



3 TOOL CALLING FLOW (WITHIN EACH SYSTEM)



4 WHY THIS DESIGN?

- 🎯 Fair comparison: same inputs, same cases
- 🛡️ Separation of concerns: extraction vs. explanation
- ⚙️ Flexibility: each system can be improved independently
- 📈 Scalability: systems can be deployed

Test Dataset — 3 Clinical Reports

Identical reports used across MedGemma 4B, 27B, and Claude — ensuring a fair, controlled comparison.



Expected : MODERATE URGENCY

Diabetes Follow-up — Alex Johnson, 52M

Plan: Increase Metformin + add Empagliflozin

- › HbA1c 8.2% (target <7%) — ABNORMAL
- › Fasting glucose 186 mg/dL — ABNORMAL
- › eGFR 62 — early diabetic nephropathy
- › BP 138/86, weight up 4 lbs



Expected: HIGH URGENCY

Chest X-Ray — Maria Rodriguez, 67F

Plan: Urgent evaluation — sputum culture — follow-up imaging in 4–6 wks

- › Right lower lobe consolidation + air bronchograms
- › Small right pleural effusion
- › Increased interstitial markings bilaterally
- › Progressive dyspnea × 10 days



Expected: LOW URGENCY

Annual Checkup — David Chen, 34M

Plan: Continue lifestyle. Flu vaccine given. Return in 1 year.

- › BP 118/74, BMI 23.2 — Normal
- › All labs within normal limits
- › Total Chol 182, LDL 98, HDL 58
- › CBC, TSH, Glucose all normal

Terminology Depth: 4B vs 27B

MedGemma 27B adds significantly more clinical specificity — but same severity reasoning.

MedGemma 4B

HbA1c 8.2% — ABNORMAL (target <7%)

eGFR 62 — ABNORMAL (normal 90+)

T2DM, HTN, Early Diabetic Nephropathy

Clinical Severity: MODERATE

MedGemma 27B

eGFR 62 mL/min/1.73m² — stage G2 CKD

HbA1c 8.2% — poor glycemic control (target <7%, 3-month avg)

Suboptimal glycemic & BP control with early end-organ damage (nephropathy)

Clinical Severity: MODERATE — same conclusion

⚠️ Key Finding: More terminology ≠ better urgency reasoning — both 4B and 27B missed the Chest X-Ray as HIGH urgency.

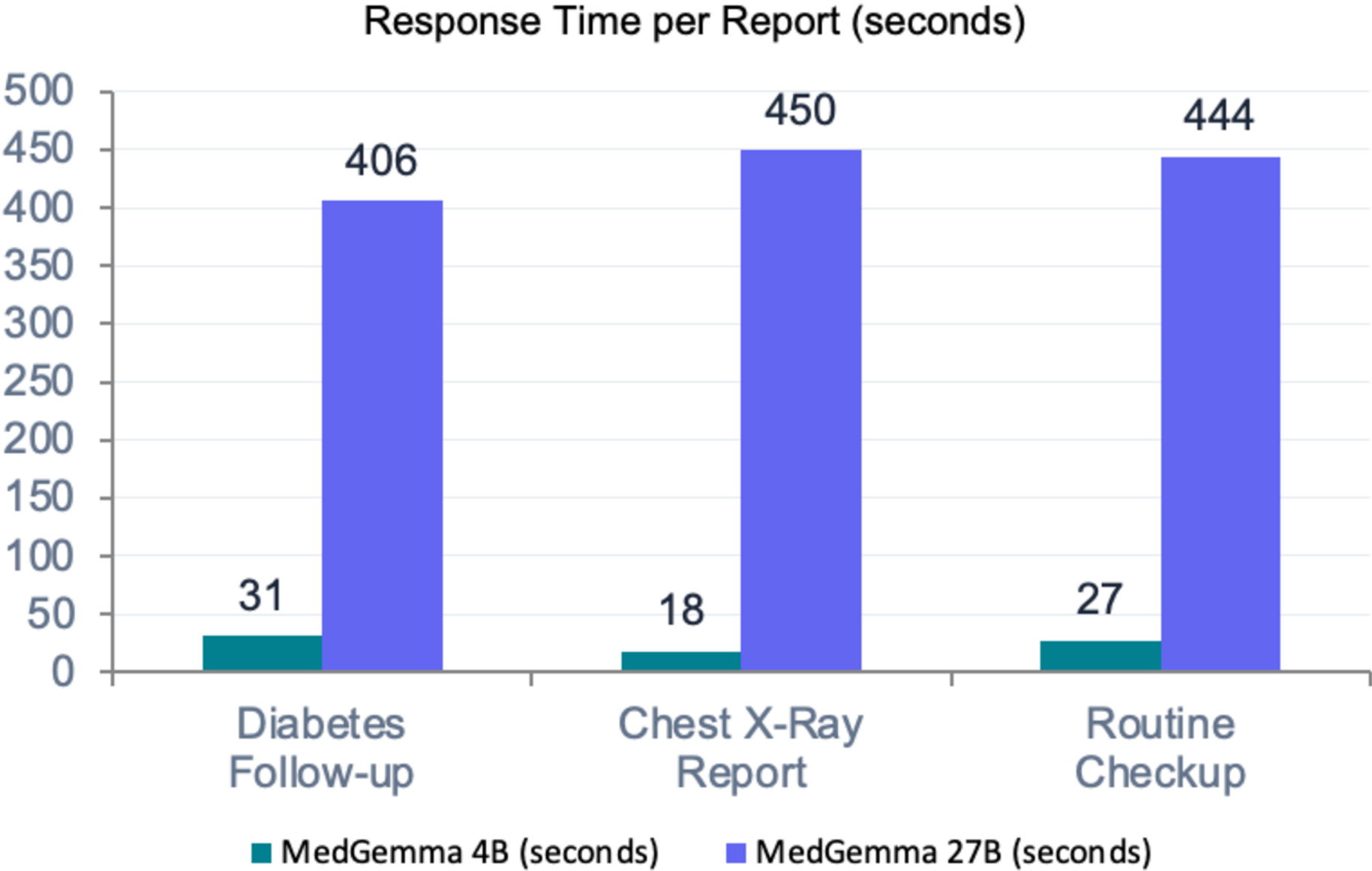
MedGemma 4B vs MedGemma 27B

Same model family — different parameter scale. How much does size matter?

MedGemma 4B	
Parameters	4 Billion
Precision	bfloat16
VRAM Required	~8.6 GB
Avg Response Time	25.2s / report
Avg Token Output	~300 tokens
Fast, efficient, production-viable. Fits comfortably on a T4 GPU. Strong clinical accuracy with concise structured output.	

MedGemma 27B	
Parameters	27 Billion (6.75× larger)
Precision	4-bit NF4 quantized
VRAM Required	~14 GB
Avg Response Time	433.1s / report
Avg Token Output	~322 tokens
Maximum medical depth. Adds reference ranges, CKD staging, parenthetical explanations. But 17.2× slower — impractical for real-time use on T4.	

4B vs 27B — Results Comparison



Report	Expected	4B Result	27B Result	Agree?
Diabetes Follow-up	Moderate	✓ Moderate	✓ Moderate	✓ Same
Chest X-Ray	HIGH	✗ Moderate	✗ Moderate	✓ Same (Both Wrong)
Routine Checkup	Low	✓ Mild	✓ Mild	✓ Same

17.2×

Slower (27B vs 4B)
433s vs 25s avg

2/3

Reports: Same
Severity Result

+23

Medical Terms Added
by 27B (total)

Why MedGemma 4B, Not 27B?

Bigger is not always better — especially in real-time clinical applications.

17.2× Faster

25s vs 433s per report on the same T4 GPU — 27B is unusable in real-time apps

Same Accuracy

Identical severity classification on 2/3 reports — scale added no reasoning gain

Same Mistake

Both missed Chest X-Ray as HIGH urgency — more parameters didn't fix the core problem

No Compression Needed

4B runs clean bfloat16 at 8.6 GB; 27B needs NF4 quantization just to fit in 14 GB VRAM

Claude Handles Communication

4B only needs to extract clinical facts accurately — which it does at 5/5 stars

More Terms ≠ Better Reasoning

27B added 23 extra medical terms but zero improvement in urgency classification

Bottom line: 4B is production-ready (25s, free, 5/5 accuracy). 27B is a research curiosity (433s, quantized, same errors).

MedGemma 4B vs Claude API

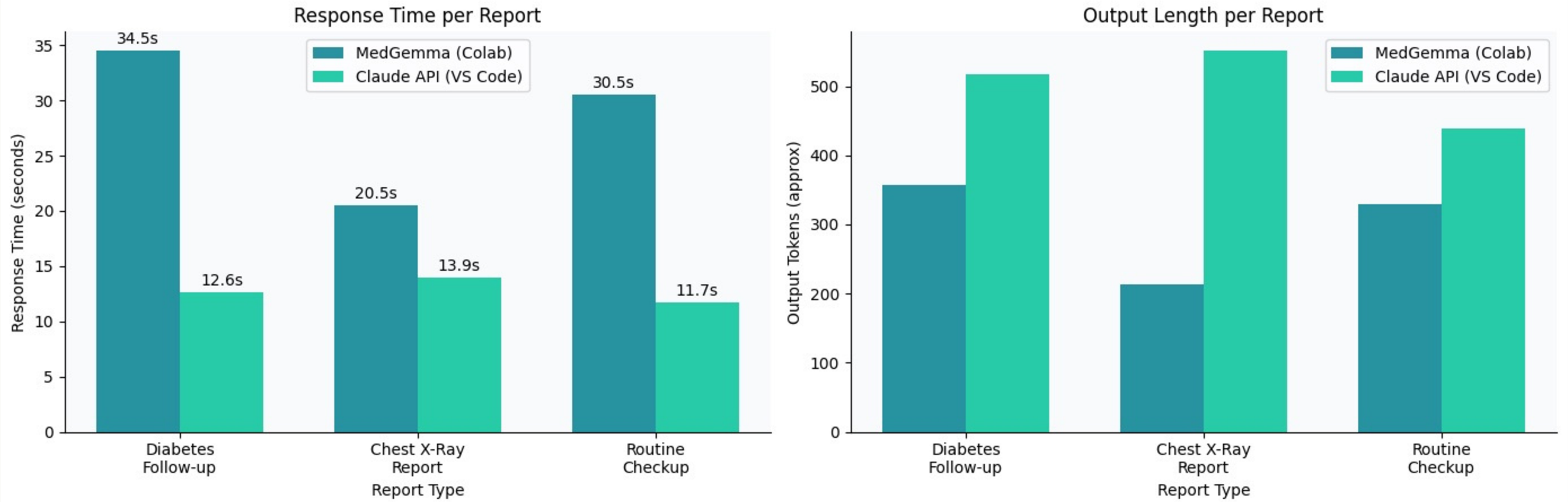
Two distinct roles — not competing, but complementary.

Dimension	MedGemma 4B (/5)	Claude API (/5)	Notes
Clinical Accuracy	★ ★ ★ ★ ★	★ ★ ★ ☆ ☆	MedGemma trained on clinical data
Medical Terminology	★ ★ ★ ★ ★	★ ★ ★ ☆ ☆	MedGemma uses ICD-10, clinical shorthand
Patient Friendliness	★ ★ ☆ ☆ ☆	★ ★ ★ ★ ★	Claude excels at plain language
Urgency Detection	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆	Claude more explicit about urgency framing
Output Completeness	★ ★ ★ ★ ☆	★ ★ ★ ★ ★	Claude produces longer structured output
Response Speed	★ ★ ☆ ☆ ☆	★ ★ ★ ★ ★	Claude ~12s vs MedGemma ~25s

MedGemma vs Claude — Performance Comparison

Response time and output length across all 3 clinical reports — based on real Colab inference results.

MedGemma (Colab) vs Claude API (VS Code) — Comparison



 **Speed Winner**

Claude 2.2×

Avg 12.7s vs 28.5s
Fastest on every report

 **Output Length**

Claude +45%

Claude ~490 tokens avg
MedGemma ~300 tokens avg

 **Biggest Gap**

Diabetes Case

34.5s (MG) vs 12.6s (Claude)
2.7× difference

 **Closest Gap**

Chest X-Ray

20.5s (MG) vs 13.9s (Claude)
1.5× difference

Output Comparison: Chest X-Ray Case

 MedGemma 4B  20.5s | 214 tokens

Key Findings:

Right lower lobe consolidation + air bronchograms
Small right pleural effusion
Increased interstitial markings bilaterally
Progressive dyspnea × 10 days

Conditions:

Right lower lobe pneumonia
Small parapneumonic effusion

Severity:

MODERATE — consolidation + effusion + dyspnea

 Claude API  13.9s

Your chest X-ray shows pneumonia in the lower right part of your lung — the infection is causing that area to fill with fluid, which is why you've been short of breath and coughing for 10 days.

There is also a small amount of fluid around your right lung (pleural effusion). This is common with pneumonia and appears small.

Pneumonia is treatable, but requires prompt medical attention — please see a doctor TODAY or TOMORROW at the latest.

What to do:

- Contact your doctor immediately
- Go to urgent care if you can't reach them
- Watch for: difficulty breathing, blue lips, chest pain, high fever

Clinical Severity: Moderate | Urgency: HIGH 

Critical Finding

Both MedGemma 4B and 27B classified the Chest X-Ray as MODERATE when the expected urgency was HIGH.

*The 27B model — 6.75× larger — made the exact same misclassification.
This proves the limitation is in reasoning strategy, not model scale.*

Why MedGemma Missed It

- › Focused on individual findings in isolation
- › Did not synthesize: patient age (67F) + 10-day trajectory + pleural effusion combination
- › Used textbook criteria — lacked contextual urgency weighting
- › Treated 'urgent clinical evaluation' recommendation as Moderate, not High

Why Claude Got It Right

- › Reasoned holistically over patient age, symptom duration, and imaging combination
- › Recognized that bilateral interstitial markings + effusion + dyspnea × 10 days = urgent
- › Applied communication-driven urgency: 'see a doctor TODAY or tomorrow'
- › Explicitly labeled: Clinical Severity: Moderate | Urgency: HIGH

Key Insights

01 The Two Models Are Complementary

MedGemma provides clinical depth (ICD-10, lab thresholds, structured findings). Claude provides communication breadth (empathy, urgency, actionable advice). Neither alone matches the dual-AI pipeline.

02 Scale ≠ Better Reasoning

MedGemma 27B (6.75× larger) uses more medical terms and adds reference ranges, But fails the same urgency classification as 4B. For patient-facing applications at 433s/report, 4B is the practical choice.

03 Claude Excels at Urgency Synthesis

Claude's superiority in urgency detection comes from holistic reasoning across patient context — not just pattern-matching on individual findings. This is the core value of the communication layer.

04 Cost-Accuracy Trade-off Is Favorable

MedGemma 4B: free, ~25s/report, strong clinical accuracy. Claude: ~\$0.002/call, ~13s, excellent communication. Combined cost per patient analysis: ~\$0.002 — highly viable for clinical deployment.

Risk, Bias & Ethical Audit

REQUIREMENT III-C

● Hallucination Risk

Both MedGemma and Claude can produce medically plausible but clinically incorrect statements. No output in this project was verified by a licensed clinician. In a medical context, incorrect information can have serious health consequences.

💡 Mitigation: Every output includes a prominent disclaimer. System is labeled educational only, not medical advice. Emergency disclaimers on all urgency assessments.

🔒 PHI & HIPAA Privacy

Real medical reports contain Protected Health Information governed by HIPAA. This prototype uses entirely fictional sample reports. Sending real PHI to external APIs (Claude API) raises serious compliance concerns.

💡 Mitigation: Production deployment requires Business Associate Agreements, data minimization, possibly on-premise model hosting to avoid transmitting PHI externally.

⚖️ Algorithmic Bias

MedGemma was trained on clinical datasets that may not represent all demographic groups equally. Clinical LLMs have been shown to perform differentially across patient populations, age groups, and medical specialties.

💡 Mitigation: Any real deployment requires fairness evaluation across demographic groups, specialty domains, and report types before release. 3-report test set is insufficient for bias detection.

📏 Model Limitations

Only 3 test reports — statistically insufficient. Urgency detection is keyword extraction from Claude's text output, which is brittle. MedGemma's multimodal capabilities (medical images) were not evaluated.

💡 Mitigation: Expand to 10+ clinician-annotated reports. Implement dedicated classification call for urgency. Add ROC-AUC curve for binary (High vs Not-High) urgency task.

Conclusion

The dual-AI architecture — MedGemma for clinical parsing + Claude for patient communication — delivers best-in-class results that neither model achieves alone.



MedGemma 4B

Best Domain Expert

ICD-10 accuracy, clinical terminology, free — production-ready at 25s/report



MedGemma 27B

Research Use Only

More terminology depth (+23 terms) but 17.2× slower — not viable for real-time apps on T4



Claude API

Best Communicator

Fastest (12.8s), highest completeness, urgency synthesis — ideal patient-facing layer



Dual-AI Pipeline

Recommended Architecture

MedGemma clinical precision + Claude patient empathy = optimal medical AI assistant

Future Work: Fine-tune MedGemma on urgency classification · Add multimodal imaging input · Expand to 10+ report types · Deploy as HIPAA-compliant app

Future Work & Improvements

What comes next to make this system more accurate, scalable, and clinically deployable.

01 Expand the Dataset

- › Scale from 3 → 100+ real clinical reports
- › Cover more specialties: cardiology, oncology, neurology
- › Include edge cases and rare diagnoses for stress testing

02 Evaluate Newer Models

- › Benchmark MedGemma 4B-IT vs Llama 3 Med, BioMistral, OpenBioLLM
- › Test GPT-4o and Gemini 1.5 Pro as Claude alternatives
- › Explore Med-PaLM 2 for radiology-specific reports

03 Fine-Tune for Urgency Classification

- › Train MedGemma on labeled urgency datasets (LOW / MODERATE / HIGH)
- › Fix the Chest X-Ray misclassification — the key gap found in this study
- › Use MIMIC-IV clinical notes as supervised training data

04 Add Multimodal Input

- › Feed actual X-ray images directly into MedGemma vision model
- › Enable MedGemma 4B-IT image understanding for radiology reports
- › Combine image + text context for richer clinical analysis

05 Build a Production-Ready App

- › Deploy as HIPAA-compliant web app with patient auth
- › Integrate with EHR systems (Epic, Cerner) via FHIR API
- › Add patient feedback loop to improve explanations over time

06 Clinical Validation

- › Partner with physicians to evaluate output accuracy
- › Run A/B test: patients with vs without AI explanation
- › Measure impact on follow-up compliance and health outcomes